



WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS5 26/05/2026 - Group 2



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Motion Suit

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



Poster

Architecture

Introduction

Results

Conclusion

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Wearable system for monitoring posture and human movements

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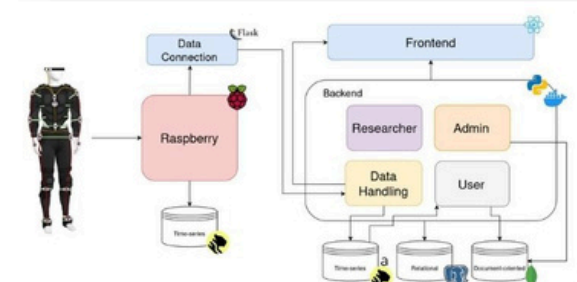
Orientador: Prof. João Almeida

Projeto em Engenharia Informática, 3ºano, LEI.

2026

This project presents a wearable suit designed to monitor human posture in real-time, providing automated ergonomic assessments and data analytics to prevent injuries in the future.

Methods



- The wearable suit integrates 14 BNO055 sensors wired through multiplexers;
- A continuous data pipeline streams movement coordinates via HTTP POST to a **centralized datastream service**;
- The **incoming data** is processed and logged into a TimescaleDB for time-series metrics and a MongoDB for document storage;
- The frontend application communicates with the backend to update a live **3D skeleton visualization**.

Fig 1 - System architecture showing the real-time data pipeline from hardware sensors to databases and frontend interfaces

Results

The platform has three different possible actors (Admin, Researcher and Suit User) that establish the full MotionSuit ecosystem.

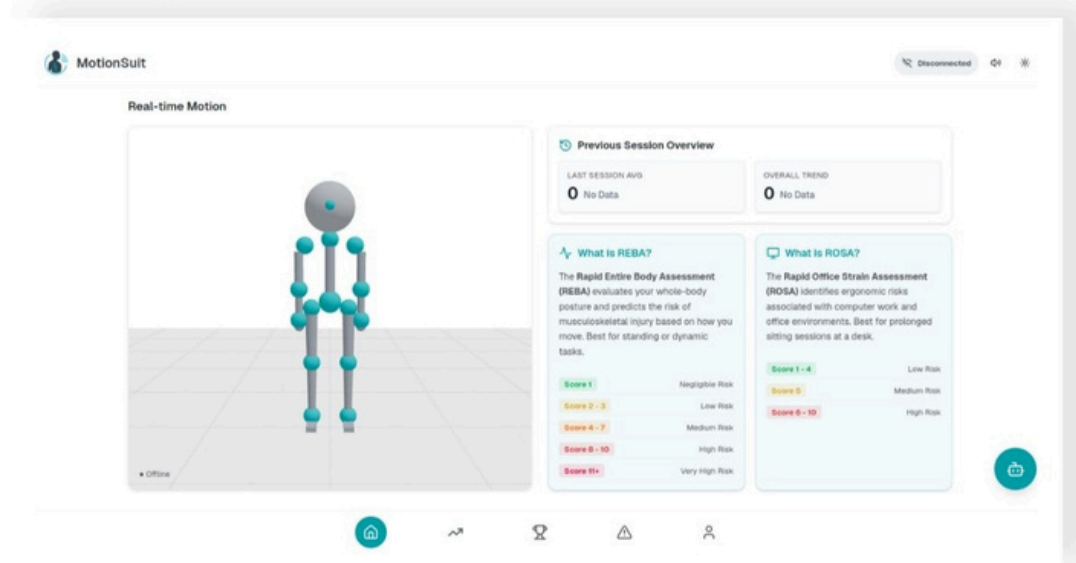


Fig 2 - Suit User dashboard displaying the live 3D skeleton tracking and automated REBA/ROSA ergonomic scores

Conclusion: The result of this work was a wearable suit, that integrates ROSA and REBA algorithms into an interactive 3D platform. It establishes a functional, scalable and effective tool for the early prevention of workplace injuries.

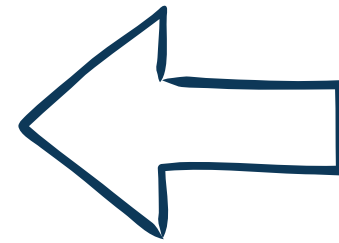
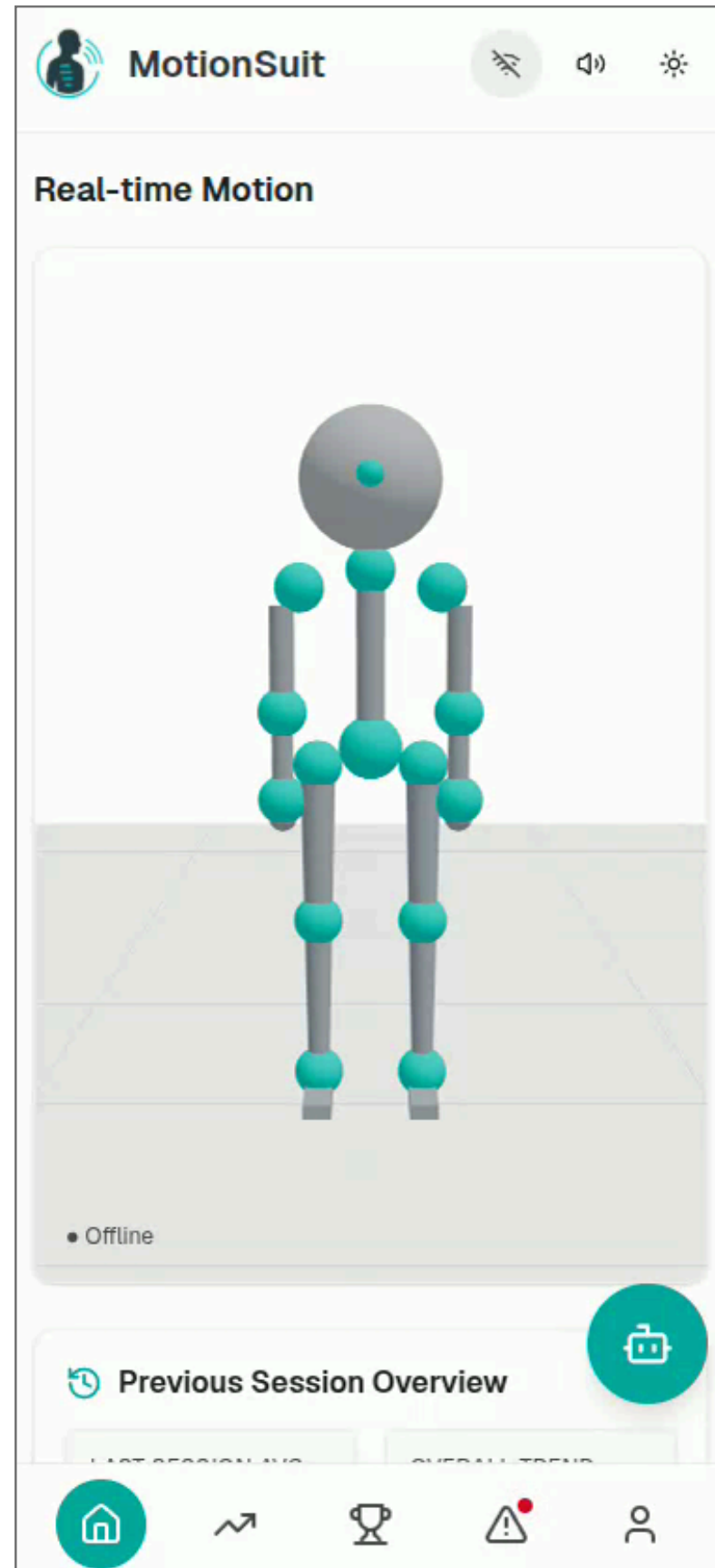


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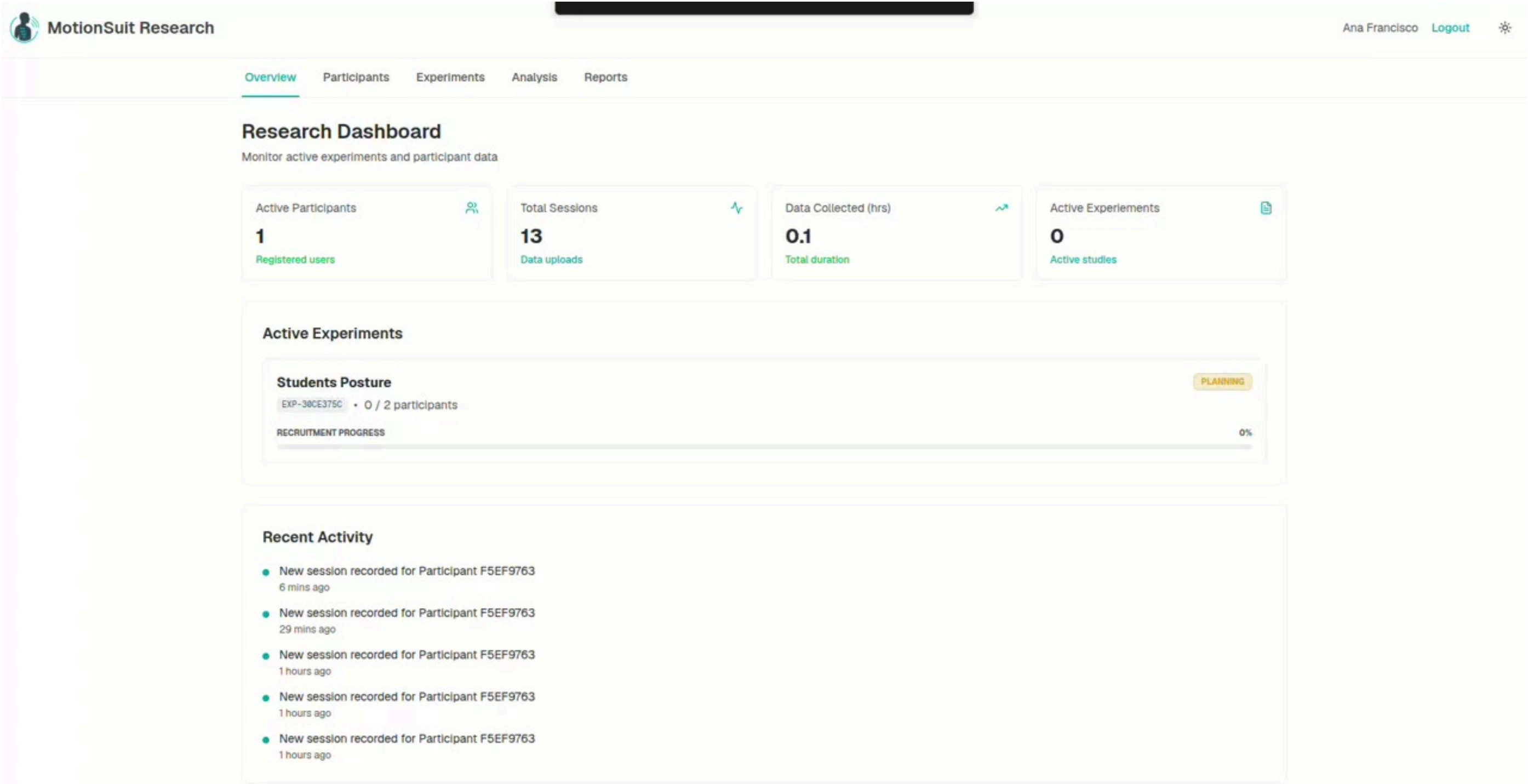
deti departamento de electrónica,
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Demo



User Functionalities

Demo



Researcher Functionalities

PROMOTIONAL VIDEO



CALENDAR



May 26

MS5: Demo + Poster for Students@DETI & Video

T-17

Done

Poster and Video



May 26

MS5: Demo + Poster for Students@DETI & Video

T-16

In Progress

Cable and Sensor Installation



<https://pei-motionsuit.github.io/microsite/>

THANK YOU!



<https://pei-motionsuit.github.io/microsite/>